Rahma Yasser Mahmoud

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PROFILE

Data Analytics instructor and educator with a strong technical foundation in SQL, Power BI, and Data Warehousing, and an academic background in Bioinformatics (Top 3 of class, GPA 3.95/4). Combine ITI Data Visualization training with hands-on teaching experience - currently a Teaching Assistant at the Faculty of Computers and Information, Assiut University, and lead Power BI instructor for an FCI Data Analysis Bootcamp. Passionate about translating complex technical concepts into clear, structured lessons, designing hands-on exercises, and mentoring students through real-world data and AI projects.

TEACHING EXPERIENCE

Power BI Instructor — Data Analysis Bootcamp 10/2025 – 01/2026
Faculty of Computers and Information (FCI), Assiut University *Assiut, Egypt*

- Designed and delivered the Power BI module of an intensive Data Analysis Bootcamp targeting third- and fourth-year FCI students.
- Covered data modelling, Power Query, DAX measures and calculated columns, dashboard design, and data storytelling.
- Built hands-on lab exercises and real-world business case studies to reinforce each concept practically.
- Mentored participants on capstone dashboard projects, providing one-on-one feedback on data modelling, visual design, and presentation.

Teaching Assistant 03/2024 – Present
Faculty of Computers and Information, Assiut University *Assiut, Egypt*

- Teach lab sessions and tutorials for undergraduate students across multiple core courses, including Data Mining, Artificial Intelligence, Machine Learning, Information Systems, and Databases.
- Prepare lab materials, hands-on exercises, and assessments aligned with course learning objectives.
- Provide one-on-one academic support during office hours and mentor students on graduation projects involving data analysis and machine learning.
- Grade assignments and exams and give written and verbal feedback to help students strengthen their technical and analytical skills.

COURSES TAUGHT

Courses delivered at university (Faculty of Computers and Information, Assiut University) and bootcamp level:

- **Power BI for Data Analysis** — FCI Data Analysis Bootcamp - 3rd & 4th year students
  Data modelling, Power Query (M), DAX measures, dashboard design, and data storytelling.
- **Data Mining** — Undergraduate - FCI, Assiut University
  Lab sessions covering data preprocessing, association rules, classification, clustering, and evaluation metrics.
- **Artificial Intelligence** — Undergraduate - FCI, Assiut University
  Tutorials and lab work on search algorithms, knowledge representation, and foundational AI techniques.
- **Machine Learning** — Undergraduate - FCI, Assiut University
  Hands-on labs on supervised and unsupervised learning, model training and evaluation, and applied ML projects in Python.
- **Information Systems** — Undergraduate - FCI, Assiut University
  Tutorials covering systems analysis and design, business processes, and the role of information systems in organizations.
- **Databases** — Undergraduate - FCI, Assiut University
  Lab sessions on relational database design, normalization, SQL, and practical query writing on real schemas.

EDUCATION

Data Visualization Track 07/2025 – 10/2025
Information Technology Institute (ITI) *Assiut, Egypt*

- Intensive professional track covering Power BI, Tableau, SQL, Data Warehousing (SSIS), and Excel.

Bachelor, Bioinformatics, Faculty of Computers and Information 09/2019 – 07/2023
Assiut University *Assiut, Egypt*

- The Third in The Class, GPA 3.95 / 4.00.
- Relevant coursework: Databases, Statistics, Machine Learning, Data Mining, Programming, Algorithms.

TECHNICAL SKILLS & TOOLS

Databases & Data Warehousing: MySQL, MS SQL Server, Oracle (Toad), SQL / PL/SQL / Analytical SQL, SSIS.

Visualization & BI: Power BI (DAX, Power Query, modeling), Tableau, MS Excel.

Programming & Scripting: Python (Pandas, scikit-learn, Keras/TensorFlow), R, Bash, Git/GitHub, Linux.

Machine Learning & AI: Classical ML (Logistic Regression, Random Forest, XGBoost), Deep Learning, model evaluation.

Teaching & Communication: Lesson planning, lab design, mentoring, technical writing, presenting in English and Arabic.

LANGUAGES

Arabic (Native) **English** (C1 — comfortable teaching and presenting)

PROJECTS

Property Finder Real Estate Data Analysis & Visualization Project
ITI Graduation Project

- Tools & Languages: Python | SQL Server | SSIS | Oracle SQL & PL/SQL | Power BI | Excel | Visual Studio.
- Built a complete BI system for real-estate analytics — from large-scale public data scraping and cleaning to warehouse design and interactive Power BI dashboards. Used as a reference example when explaining the full BI lifecycle to students.

Customer Churn Rate Analysis Dashboard

- Tools & Languages: Power BI.
- Power BI dashboard analyzing customer behavior and churn patterns to support data-driven decisions; reused as a teaching case for DAX measures and storytelling.

Sales Performance Analysis for Electronics Retail Dataset

- Tools & Languages: Analytical SQL (Aggregations, Window Functions, Ranking, Pivoting).
- Evaluated sales performance across regions, product categories, and salespersons using analytical SQL; reused as a lab exercise for SQL students.

Glaucoma Prediction using Deep Learning
Bachelor Graduation Project

- Tools & Languages: Django, PostgreSQL, Python, Keras, TensorFlow, HTML, CSS, JS.
- Built a website to help ophthalmologists predict glaucoma from fundus images, reaching ~90% validation accuracy.

LUAD Gene Prioritization Project

- Modular Python pipeline integrating TCGA tumor RNA-seq and GTEx normal tissue data; demonstrates classical ML (Logistic Regression, Random Forest, XGBoost) on real biomedical data.